Fixed Income Securities – Valuation, Pricing, Volatility

Basics of Time Value of Money

Compounding

Discounting

Basics of Time Value of Money

- Discounting
- Compounding
- Single Cash Flow
  - PVIF
  - FVIF
- Annuity
  - PVIFA
  - FVIFA
- Refer Excel file “2.1.FIS” → Sheet “TVM”

Pricing a Vanilla Bond

Different coupon same rates

Convexity

Pricing different bonds at same discount rate

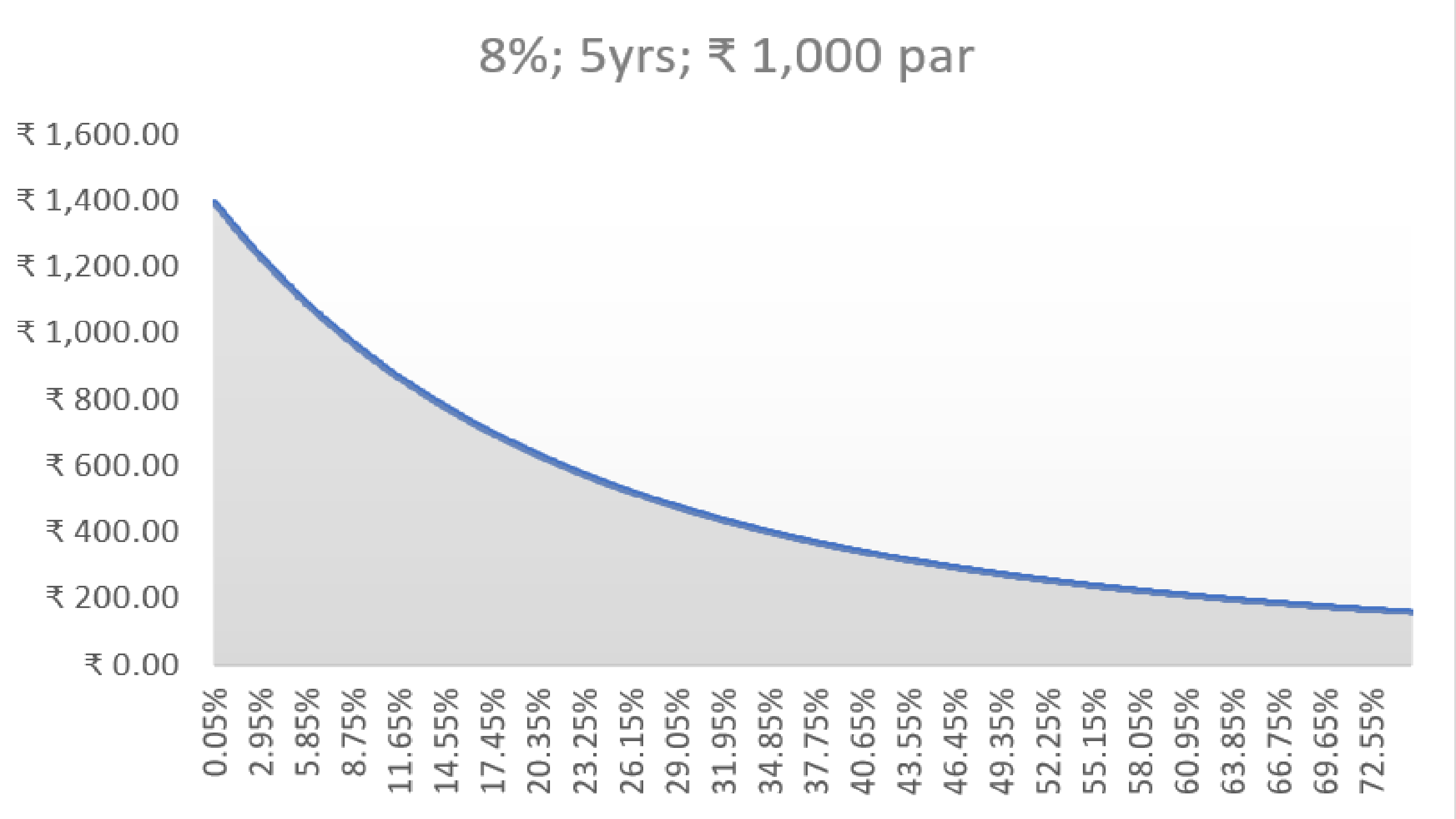
- Par value of bond maturing in 5years is Rs.1,000 maturing at par with a discount rate of 6%.
- Consider three different bonds with coupon rates payable once as:
  - 4%
  - 6%
  - 8%
- Coupon < Rate → deficiency → trading below par → discount
- Coupon > Rate → excess → trading above par → premium
- Coupon = Rate → *na* → trading at par → *na*
- Refer Excel file "2.1.FIS" → Sheet "1."

Discounting deficiency/excess

- If coupon is above discount rate … the difference in excess
- If coupon is below discount rate … the difference in deficient
- Calculate the amount of deficiency or excess per 100 of par value in the excel file
- Refer Excel file “2.1.FIS” → Sheet “2.”

Same Coupon → same maturity → different rates

- Refer Excel file "2.1.FIS" → Sheet "3."
- Calculate Bond price with following rates:
  - 7%
  - 8%
  - 9%
- Observe the changes in bond prices from 8% → 7% & 8% → 9%
- Percentage change and value changes are not same


8%; 5yrs; ₹ 1,000 par
₹ 1,600.00
₹ 1,400.00
₹ 1,200.00
₹ 1,000.00
₹ 800.00
₹ 600.00
₹ 400.00
₹ 200.00
₹ 0.00
0.05%
2.95%
5.85%
8.75%
11.65%
14.55%
17.45%
20.35%
23.25%
26.15%
29.05%
31.95%
34.85%
37.75%
40.65%
43.55%
46.45%
49.35%
52.25%
55.15%
58.05%
60.95%
63.85%
66.75%
69.65%
72.55%

Convexity

- Calculate bond price for 49 different rates.
- Keep all the factors similar to previous bond
- Start the discount rates from 1% and increase rates by 50 basis points up to 25%
- Plot the price and rates in a graph.
- Observe the price line
- This is referred to as convex to the origin

Pricing a Vanilla Bond

Coupon

Time to Maturity

Rates

Different Coupon → same Maturity → Different rates

- Refer Excel file "2.1.FIS" → Sheet "4."
- A Rs.1,000 par value bond retiring after 5 years repayable at par.
- Estimate Bond price with following different coupons → 4%, 6% 8%, 10% and 12%
- Discount rate is assumed as 8%
- Changes discount rates by ±1% for each five cases
- Which bonds are more sensitive to interest rates?
- Estimate percentage price change for each cases.

Same Coupon → Different Maturity → Different rates

- Refer Excel file "2.1.FIS" → Sheet "5."
- A Rs.1,000 par value bond with 8%pa coupon payable annually.
- Maturities (in years) for different bonds are – 3, 5, 7, 10, 15, 20, 30 years each.
- Discount rate is assumed as 8%
- Changes discount rates by ±1%
- Estimate absolute price change and percentage price change
- Identify which bond is more sensitive to interest rates changes?

Different Coupon → Different Maturity → Different rates

- Refer Excel file "2.1.FIS" → Sheet "6."
- An investor is considering the nine different bonds with annual coupon payments (if any)
- Identify the relationships.
- Change the rates by ± 1.0% and identify the relationships
- Identify the highest and lowest price changes in percentages

Estimating the Yields … YTM

Yield to Maturity (YTM)

- It is Internal Rate of Return of a bond
- the sum of the present values equals the price of the bond.
- It is the *implied* market discount rate.
- The three assumptions:
  - The investor holds the bond till maturity
  - Investor receives all coupon and principal payments in full on scheduled dates – no default
    - This assumptions makes YTM as *promised* yield – yield if issuer does not default
  - The investor (bond holder) reinvests the coupons at the same yields

Yield to Maturity (YTM)

- The formula

$$Bond\ Market\ Price = \frac{C}{(1+r)^1} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \cdots + \frac{C+R}{(1+r)^n}$$

- C = Coupon
- r = YTM
- R = Redemption price
- n = Tenor, time to maturity of bond
- It can be estimated by trial-and-error method.

Yield to Maturity (YTM)

- A four year, 5% annual coupon payment bond is priced at Rs.1050 per Rs. 1000 of par value.
- Estimate ytm?
- $1{,}050 = \frac{50}{(1+r)^1} + \frac{50}{(1+r)^2} + \frac{50}{(1+r)^3} + \frac{50+1{,}000}{(1+r)^4}$
- Is it trading at premium or discount?
  - Premium
- So, YTM is above or below Coupon rate?
  - Below
- Give the closest YTM …

Solve the following …

- A five-year 7% annual coupon payment bond is priced at Rs.1,063 per Rs 1,000 par value.
- *Use Trial-and-error*
- US Institutional Investor owns a six-year, 2.5% annual payment bond having a par value of $10 million. The bond currently is priced at $9,817,567.70
- Solve Questions in Excel file “2.1.FIS” → Sheet “7.”
- Use trial-and-error
- Is there any other way in excel?

Constant Price Trajectory

- Change in the price of a fixed-income bond over time.
- This trajectory shows the “pull to par” effect on the price of a bond trading at a premium or a discount to par value.
- A bond with a par value of Rs.1,000 with a coupon rate of 8% payable annually redeemable at par after 5years.
- Identify the bonds price trajectory for the life of the bond with discount rate of 6%, 8% and 10%.
- Plot a graph and discuss its relationship. Refer Excel file “2.1.FIS” → Sheet “9.”

Relationships between bond characteristics

- Inverse Effect
  - market discount rate increases, the bond price decreases
- Convexity Effect
  - Price change is higher when interest rate falls compared when interest rate rises.
- Coupon Effect
  - Low coupon bond is more sensitive to interest rates change compared to high coupon bond
- Maturity Effect
  - High maturity will have high price change